Overview
Shepherd

Kairo Autonomi's Shepherd application is the software on the Operator Control Unit (OCU) that allows the human operator to Tele-Operate or oversee autonomous playback operations. This document introduces the basic operation of Shepherd.

Shepherd serves as the OCU interface for several different communication protocol packages and robotic control executables. This document applies to all validated configurations. If you are interested in using a different communication protocol or robotic control executable contact Kairos Autonomi. Shepherd has been validated to work with:

- djSharedLink and IVN
- JAUS configured IVN
- djSharedLink, djDRIVEWB, and djMimic
- JAUS configured VAK (ASI's Mobius Vehicle Control Unit executable)

Shepherd supports several different modes of operation, which modes are dependent on the version of Shepherd:

- Manual (all versions)
- Tele-Op (all versions)
- Single Path Playback (Shepherd with Pathing only)
- Multiple Path Playback (Shepherd with Pathing only)

Interface Overview

The Shepherd interface has two primary tabs: “Asset View”, and “Configuration”. The Asset View provides an overview and controls for the operation of a single robot. The Configuration tab provides developmental and installation level configuration options. This document covers only the Asset View.

The “Asset View” tab has four sub-areas: asset access, asset feedback, views, and operational tabs.

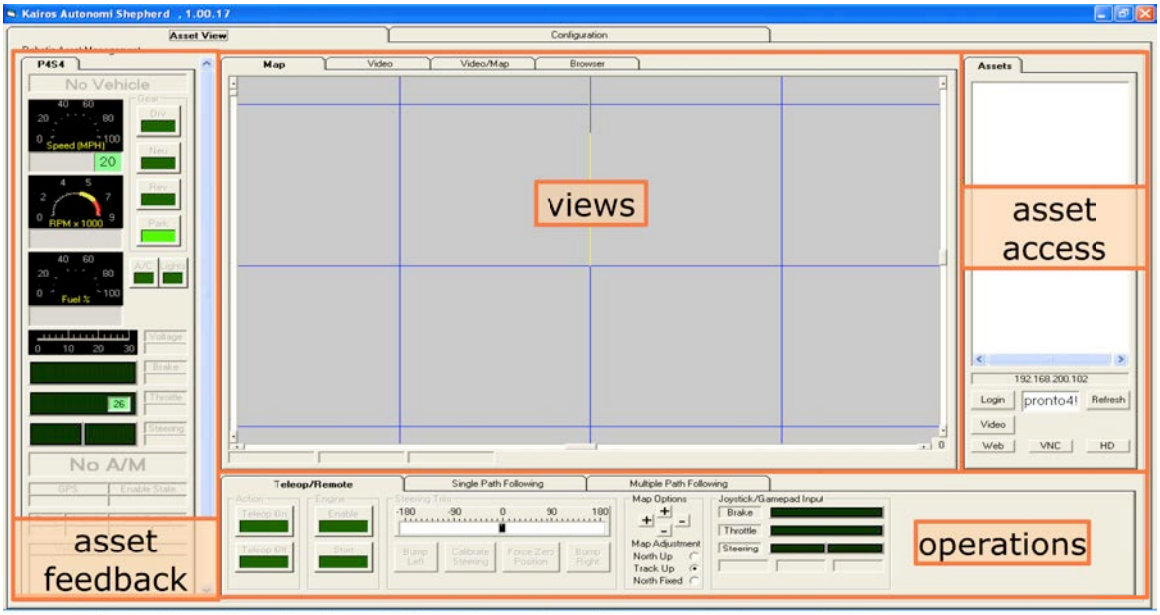

Figure A-1: Shepherd's Asset View layout, with no robots available

Asset Access Tabs

On the right side of Shepherd's main window are tabs related to available assets and their communication.

Assets

The "Assets" tab, on the right-hand side of Shepherd, allows a user to see each asset that is currently available, including vehicles (VEH), radio relays (RLY), cameras (CAM), and other OCUs (OCU). Each asset's entry can be expanded to get additional information.

Each asset type has sets of possible functionality. For example, a vehicle asset will have path recording and playback options, but a camera asset will not. Shepherd displays controls for an asset's possible functionality.

Active robotic assets are highlighted with yellow text on a grey background. Non-robotic and inactive robotic assets display with black text on a white background. The total number of primary assets is identified immediately above the assets list. When an asset is selected, its text will be white on a blue background.

Figure A-2: Assets tab

To gain or relinquish control of a robot, the robot's entry must be selected, then the Login/Logout button clicked.

Cameras associated with another asset can be accessed independently of the asset, by selecting the asset, then clicking the Video button.

The computers of other assets may be accessed by selecting the asset, then clicking the VNC button.

Computer files on other assets may be accessed by selecting the asset, then clicking the HD button.

To refresh the list of available assets, click the Refresh button.

Status

The "Status" tab, on the right-hand side of Shepherd, becomes available when asset control is activated. If there is no robot logged into the OCU, there will *not* be a "Status" tab. This tab displays a system status message log and basic communications log, along with a Logout button.

Figure A-3: Status tab, after a robot (with the I/P 192.168.200.103) has been logged in

Asset Feedback Tabs

On the left side of Shepherd's main window are tabs for each active asset.

P4S4 Tab

The "P4S4" tab, on the left hand side of Shepherd, shows the state of the active robot and provides graphical user interface (GUI) control of some auxiliary functions and gears.

The vehicle currently being commanded is shown at the top. On the left side are gauges and numerical displays showing speed, RPM, fuel level, battery voltage, brake percentage, throttle percentage, and steering degrees. In the lower right of the speed gauge and the right side of the throttle gauge are the maximum setting for those controls (e.g., in Figure A-4 the vehicle could not be commanded to exceed 20 MPH or 26% of the throttle actuator's travel).

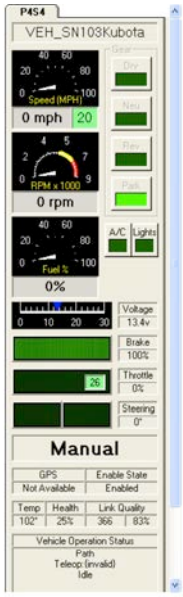

Figure A-4: P4S4 Tab, with "VEH_SN103Kubota" logged in

On the right side are gear and axillary function indicators/buttons. These button's backgrounds may turn yellow when the asset is shifting out of the given gear (e.g., shifting from reverse to park, the "Rev" button will have a yellow background until the vehicle is in park). To change gears with these buttons the system *must* be in Tele-Op mode *and* the Deadman button engaged. The vehicle's heating and ventilation controls, as well as the headlights, can be turned on and off with the "A/C" and "Lights" buttons, respectively.

The bottom shows additional system statuses, including whether the system is in "Auto" or "Manual" mode, GPS status, GPS satellite count, enabled state, system temperature, and link quality metrics.

In the system status area, if the "Auto"/"Manual" and "Link Quality" fields have a red background, this indicates either a loss of communication or the asset is busy (e.g., active in another OCU). The numerical value in "Link Quality" shows either an active packet count or, in the case of a lost link, the length of the outage in seconds.

View Tabs

In the top center of Shepherd's main window are tabs providing various different views of the currently operational vehicle and additional documentation. There are four main tabs: Map, Video, Video/Map, and Browser. Other tabs may appear when assets have additional functionally, such as multiple or omni-camera views.

Map

The "Map" tab shows the robot's orientation in relation to loaded paths or recently driven waypoints.

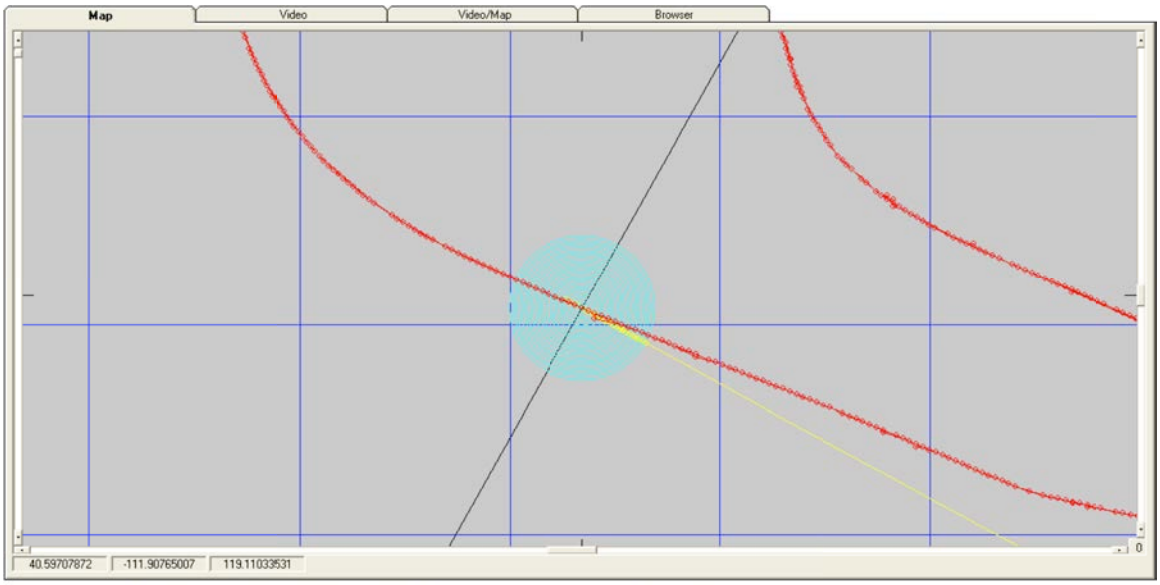

Figure A-5: Map tab, with vehicle on prerecorded path

In the lower-left corner of the tab window are three values representing latitude, longitude, and heading. If these values are not present or highlighted in red, check all GPS cable connections and possible interference (e.g., robot is inside a garage).

Map controls are available on the "Teleop/Remote" tab in the operations area.

Video

The "Video" tab, shows one primary and up to three auxiliary video feeds. If programmed, the top of the video feed will include the robot's short name and relative direction of the camera. Double-clicking an auxiliary feed will cause it to become the primary video feed.

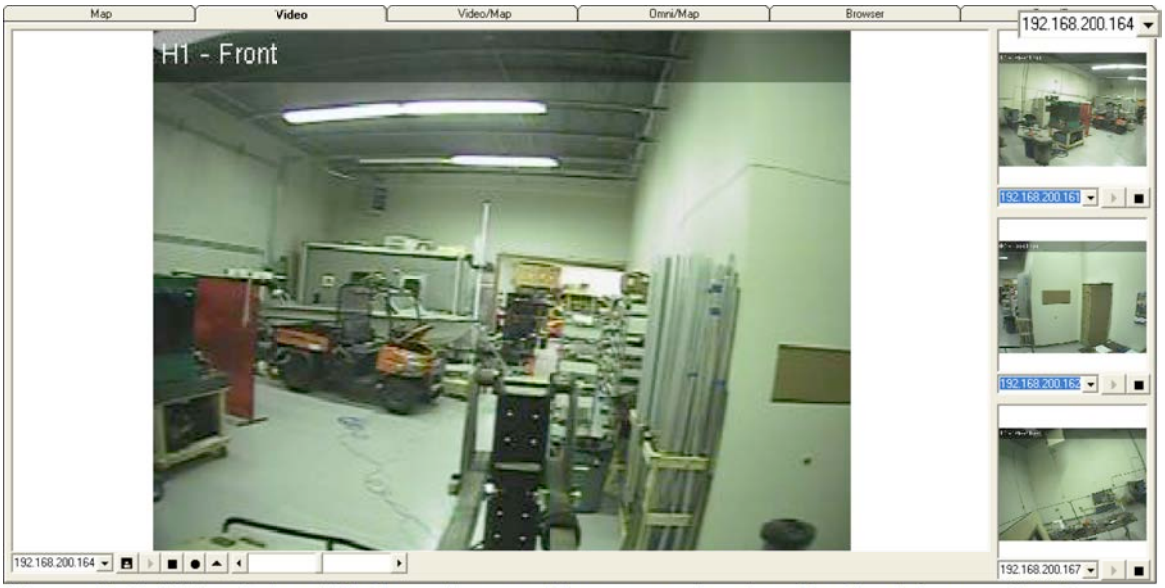

Figure A-6: Video tab, with the primary video area showing the feed from a front facing camera on an H1 vehicle, that is sitting in the Kairos development area.

In the lower left corner of each video feed area is a drop-down selector with a list of cameras with active video feeds from all assets on the network. To change the source for a specific video feed area, simply click the drop-down, click the desired camera, then click the play button.

To the immediate right of the camera feed selector, are a set of buttons that control the recording and playback of videos. The primary video feed display has all six (6) buttons. Auxiliary video feed displays have only the play and stop buttons. Each button performs the following function:

- Screen Shot – Save a static screen shot of the main video window.
- Play – Stream selected video feed.
- Stop – Stop the recording of the current video feed ***or*** the streaming video.
- Start Recording – Start recording the current video feed to '`C:/GC07_vids/`'
- Maximize – Maximize the video to full screen. Press the 'ESC' key to return to the normal interface.

To the immediate right of the primary video recording and playback buttons is a slider control. This controls the relative width of the primary and auxiliary video windows.

Video/Map

The "Video/Map" tab provides a reduced version of the Map tab's content and the primary video feed from the Video tab. For more information refer to the previous two sections.

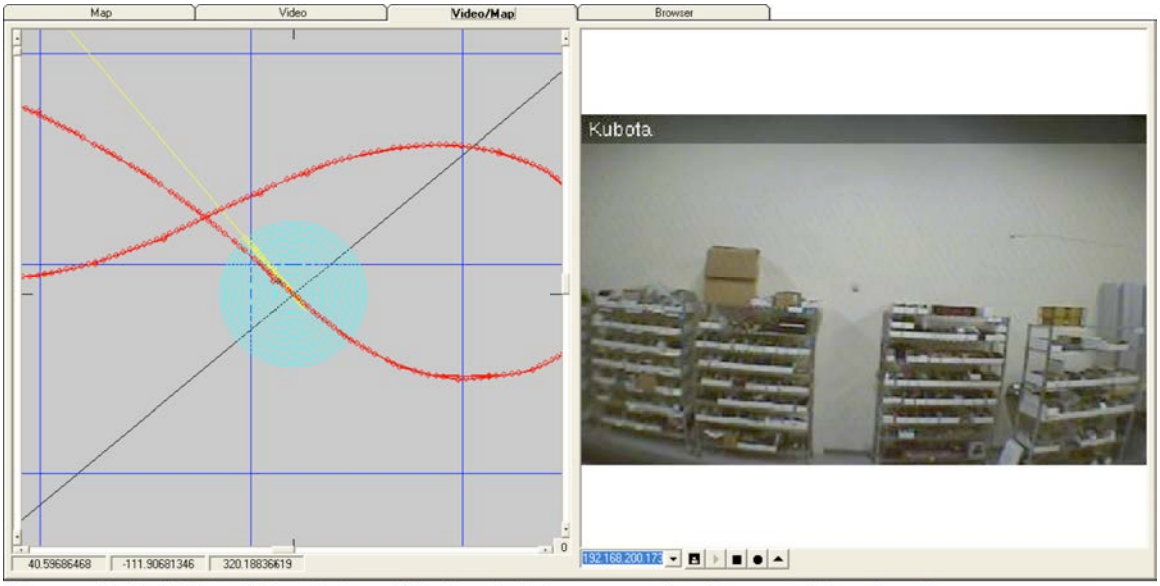

Figure A-7: Video/Map tab, with the joint map and primary video feed.

Browser

The "Browser" tab provides alternative access to operational data and reference documentation accessible directly from the asset. The default "Docs" tab provides access to more detailed documentation for the active asset.

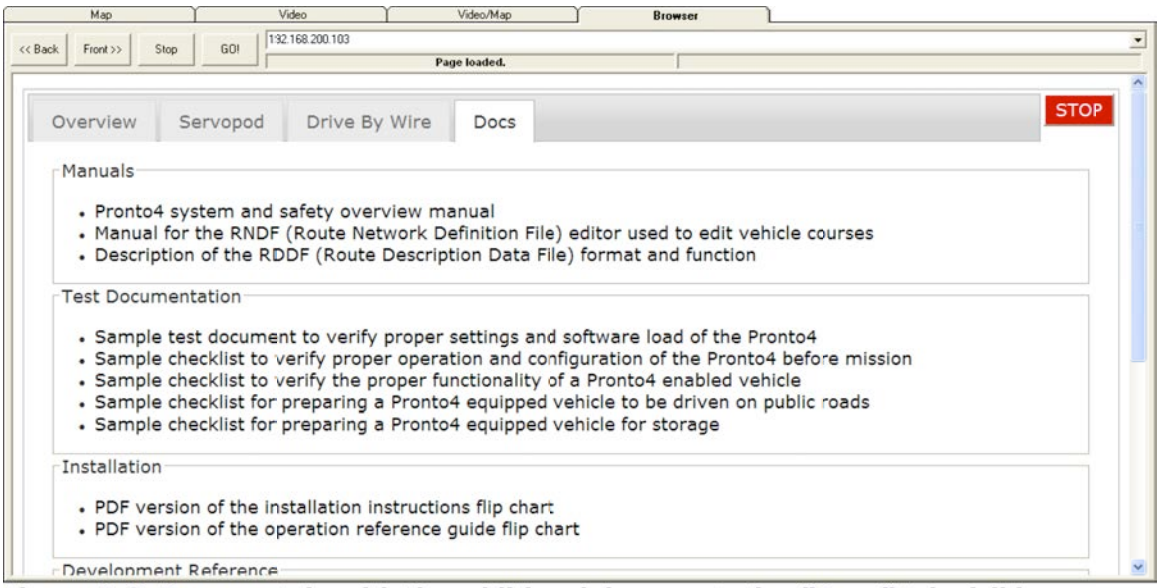

Figure A-8: Browser tab, with the additional documentation "Docs" tab visible.

Operational Tabs

Most robotic operations are performed from the operational tabs. The available tabs are dependent on the version of Shepherd. The primary operational tabs are: Tele-Operation, Single Path Following, and Multiple Path Following.

Teleop/Remote

The "Teleop/Remote" tab is the primary set of controls to enable tele-operations, calibrate steering, and observe recognized tele-operational commands.

Figure A-9: Teleop/Remote tab, with active Joystick controls.

The "Action" buttons indicate and control whether the system is in Tele-Operational mode.

NOTES
• When a robot is in Tele-Op mode, the Logout button on the "Assets" and "Status" tabs is inactive. The Teleop Off button must be clicked to reactivate the Logout functionality. • When the deadman button is depressed, the Teleop On button is inactive.

The "Engine" buttons indicate and control the enable and vehicle engine start functions.

The "Steering Trim" buttons control the relationship between the tele-op controller's steering and the vehicle's steering. When the vehicle is initially, manually positioned, the vehicle steering wheel should be centered and the vehicle wheels should be straight. At this point the OCU operator clicks the Calibrate Steering button, to set the baseline steering relationship. During operations there may be conditions that cause the robot to pull in one direction (e.g., damage); when such conditions are encountered the Bump Left and Bump Right buttons can be used to temporarily reset the steering relationship. Adjustments done with the bump buttons are not designed to be maintained for the remainder of a robotic operation. The ForceZero Position button can be used to retain steering bump adjustments for the remainder of the current operation.

The "Map Options" area controls the view of the map in both the "Map" and the "Video/Map" tabs.

The +/- buttons at the top of the Map Options area move the background image (e.g., Google Map Image) in the corresponding directions. The optional selection buttons adjust the map orientation as follows:

North Up — The vehicle remains centered in the view, the map remains oriented with North toward the top of the window, and the map is moved as the vehicle follows its path.

Track Up — [Default] The vehicle remains centered in the view and oriented toward the top of the window, the map rotates and is moved as the vehicle follows its path.

North Fixed — The map remains centered and oriented with North toward the top of the window, and the vehicle marker moves as the vehicle follows its path.

The "Joystick/Gamepad Input" area provides gauges and textual indicators of current controller commands. Immediately below the "Steering" gauge's label is a field that shows either "Deadman" to indicate that the Deadman is not being being applied or "MotionOn" to indicate the system is in Tele-Op mode and the Deadman is being applied.

Single Path Following

The "Single Path Following" tab (only available in versions of Shepherd with Pathing) provides the controls to record and playback a single path recording.

Figure A-10: Single Path Following tab, with no path selected.

The top of the Single Path Following tab displays three options.

"Drive @" with the options "Recorded Speed" or "Fixed Speed". This indicates whether the path should be played back at the recorded speed or a single, fixed speed.

"Path:" is the currently loaded path file.

"@"/ "fixed mph" is the single, fixed speed to playback the loaded path.

The bottom of the Single Path Following tab displays buttons that perform the functions listed in Table A-1.

Table A-1: Single Path Course buttons

	Select path	Select the path and file name of a previous recording for playback. Folder outline turns blue when a path is loaded for playback.
	Play	Start playing back a previously recorded path ***OR*** Start recording a new path. Triangle is black when system has path for playback ***or*** is prepared to record a new path.
	Pause	Pause current playback ***OR*** Pause recording.
	Stop	Stop playback of a previously recorded path ***OR*** Stop the recording of the current path. Recordings are automatically saved to `\\###.###.###.###\c\GC07_rddf\`, where `###.###.###.###` is the robot's IP address.
	Prepare Recording	Prepare the system to record a new path. Circle turns red when a recording is prepared.
	Add Attribute	Manually add a playback attribute, such as playback in Reverse.

Attributes ***must*** be set during the recording of a path. For example, if the vehicle's manual driver puts the vehicle into reverse, then the OCU operator will need to add the Reverse attribute at the same time.

Multiple Path Following

The "Multiple Path Following" tab (only available in versions of Shepherd with Pathing) provides the controls to playback a series of 1-3 previously recorded paths.

Figure A-11: Multiple Path Following tab, with no paths selected.

The top of the Multiple Path Following tab displays three options.

"Drive @" with the options "Recorded Speed" or "Fixed Speed". This indicates whether the path should be played back at the recorded speed or a single, fixed speed.

"Mission:" is the currently loaded mission file. There are three corresponding buttons:

Edit – Allow the currently loaded mission file to be edited.

Save – Save the currently loaded mission settings.

Load – Load a previously saved mission file.

"#"/"infinite"/"loop cycles" is the number of times the Loop path should be played back before progressing to the Egress path. If left empty, then the Loop path will playback until commanded to do otherwise.

The bottom of the Multiple Path Following tab displays buttons that perform the following functions:

Table A-2: Multiple Path Course buttons

Ingress>	Ingress>	Select an initial path.
Loop>	Loop>	Select a path that will be run continuously until the Stop button is clicked. If an Ingress path is selected, then the Loop will start automatically when the Ingress path playback is complete.
Egress>	Egress>	Select an exit path.
▶	Play	Start playing back a previously recorded path or mission set.
■	Stop	Stop playback of the current mission. If running a Loop path, this will stop the looping playback; when the play button is pressed the mission will complete the current loop then playback the Egress path.

The buttons to the right of the Ingress>, Loop>, and Egress> buttons are text fields displaying the selected path and path playback velocity. The paths can be directly entered and edited in the selected path's text field. The velocity field is in miles per hour (MPH).

NOTE

If the speed for either ingress or egress are set to zero (i.e., "0 fixed mph") then the respective path will be not be played.

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